

Table S4: Gene ontologies enriched in Ning *et al* dataset differentiating GOLD 0 from GOLD2 smokers.

GOID	Ontology	Term	p
GO:0006414	biological_process	translational elongation	1.43E-24
GO:0006412	biological_process	translation	2.89E-19
GO:0043284	biological_process	biopolymer biosynthetic process	1.11E-11
GO:0034961	biological_process	cellular biopolymer biosynthetic process	3.21E-11
GO:0009059	biological_process	macromolecule biosynthetic process	1.01E-10
GO:0009058	biological_process	biosynthetic process	3.86E-10
GO:0034645	biological_process	cellular macromolecule biosynthetic process	2.67E-09
GO:0044249	biological_process	cellular biosynthetic process	4.75E-09
GO:0009987	biological_process	cellular process	5.78E-09
GO:0010467	biological_process	gene expression	7.06E-09
GO:0044237	biological_process	cellular metabolic process	7.60E-08
GO:0008152	biological_process	metabolic process	2.18E-07
GO:0043170	biological_process	macromolecule metabolic process	2.18E-07
GO:0044260	biological_process	cellular macromolecule metabolic process	1.44E-06
GO:0044238	biological_process	primary metabolic process	1.77E-06
GO:0019538	biological_process	protein metabolic process	2.47E-06
GO:0044267	biological_process	cellular protein metabolic process	7.74E-06
GO:0010035	biological_process	response to inorganic substance	8.88E-06
GO:0043283	biological_process	biopolymer metabolic process	3.69E-05
GO:0006172	biological_process	ADP biosynthetic process	4.18E-05
GO:0009136	biological_process	purine nucleoside diphosphate biosynthetic process	4.18E-05
GO:0009180	biological_process	purine ribonucleoside diphosphate biosynthetic process	4.18E-05
GO:0009188	biological_process	ribonucleoside diphosphate biosynthetic process	4.18E-05
GO:0016043	biological_process	cellular component organization	4.79E-05
GO:0042274	biological_process	ribosomal small subunit biogenesis	4.99E-05
GO:0044085	biological_process	cellular component biogenesis	7.38E-05
GO:0046031	biological_process	ADP metabolic process	0.000116

GO:0034960	biological_process	cellular biopolymer metabolic process	0.000173
GO:0009133	biological_process	nucleoside diphosphate biosynthetic process	0.000256
GO:0009135	biological_process	purine nucleoside diphosphate metabolic process	0.000256
GO:0009179	biological_process	purine ribonucleoside diphosphate metabolic process	0.000256
GO:0030155	biological_process	regulation of cell adhesion	0.000383
GO:0022613	biological_process	ribonucleoprotein complex biogenesis	0.000383
GO:0016192	biological_process	vesicle-mediated transport	0.00079
GO:0033003	biological_process	regulation of mast cell activation	0.000823
GO:0009185	biological_process	ribonucleoside diphosphate metabolic process	0.000823
GO:0060325	biological_process	face morphogenesis	0.000823
GO:0002886	biological_process	regulation of myeloid leukocyte mediated immunity	0.000823
GO:0044092	biological_process	negative regulation of molecular function	0.00111
GO:0006364	biological_process	rRNA processing	0.00111
GO:0042273	biological_process	ribosomal large subunit biogenesis	0.001302
GO:0002697	biological_process	regulation of immune effector process	0.001398
GO:0016072	biological_process	rRNA metabolic process	0.001504
GO:0042254	biological_process	ribosome biogenesis	0.001709
GO:0048545	biological_process	response to steroid hormone stimulus	0.001741
GO:0000302	biological_process	response to reactive oxygen species	0.001791
GO:0010926	biological_process	anatomical structure formation	0.001797
GO:0030260	biological_process	entry into host cell	0.001803
GO:0044409	biological_process	entry into host	0.001803
GO:0046718	biological_process	entry of virus into host cell	0.001803
GO:0051806	biological_process	entry into cell of other organism during symbiotic interaction	0.001803
GO:0051828	biological_process	entry into other organism during symbiotic interaction	0.001803
GO:0052126	biological_process	movement in host environment	0.001803
GO:0052192	biological_process	movement in environment of other organism during symbiotic interaction	0.001803

GO:0009725	biological_process	response to hormone stimulus	0.001803
GO:0060323	biological_process	head morphogenesis	0.001803
GO:0060324	biological_process	face development	0.001803
GO:0051179	biological_process	localization	0.002248
GO:0010033	biological_process	response to organic substance	0.002379
GO:0042221	biological_process	response to chemical stimulus	0.00244
GO:0007162	biological_process	negative regulation of cell adhesion	0.002479
GO:0016044	biological_process	membrane organization	0.002644
GO:0010038	biological_process	response to metal ion	0.002981
GO:0002246	biological_process	healing during inflammatory response	0.00383
GO:0032964	biological_process	collagen biosynthetic process	0.00383
GO:0006787	biological_process	porphyrin catabolic process	0.00383
GO:0033015	biological_process	tetrapyrrole catabolic process	0.00383
GO:0044275	biological_process	cellular carbohydrate catabolic process	0.004204
GO:0032410	biological_process	negative regulation of transporter activity	0.004771
GO:0010171	biological_process	body morphogenesis	0.004771
GO:0060322	biological_process	head development	0.004771
GO:0009719	biological_process	response to endogenous stimulus	0.005017
GO:0009605	biological_process	response to external stimulus	0.00507
GO:0006091	biological_process	generation of precursor metabolites and energy	0.00538
GO:0006200	biological_process	ATP catabolic process	0.00623
GO:0006901	biological_process	vesicle coating	0.00623
GO:0009612	biological_process	response to mechanical stimulus	0.00664
GO:0019062	biological_process	virion attachment to host cell surface receptor	0.007026
GO:0030198	biological_process	extracellular matrix organization	0.007472
GO:0030837	biological_process	negative regulation of actin filament polymerization	0.007562
GO:0009132	biological_process	nucleoside diphosphate metabolic process	0.007963
GO:0006900	biological_process	membrane budding	0.007963
GO:0030833	biological_process	regulation of actin filament polymerization	0.008187
GO:0032272	biological_process	negative regulation of protein polymerization	0.008732
GO:0001568	biological_process	blood vessel development	0.009191